13/10/15 Activity 5 : Normality Detection using Q-Q plots

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Part A: Intepret four Q-Q plots

A Q-Q plot compares samples data with what we would expect from a normal distribution

Normal: points stay close to straight reference line

Right skewed: points typically curve away from line, with a longer upper tail

left skewed: Show opposite curvature.

Heavy tailed: The tails extend farther than expected

light tailed: The tails are less extreme

Outlies: One or more points lie unusually far from overall pattern.

Chart 1: Daily study time, n=60

Normal: no

Shape: Bimodal

Evidence: The points form two clusters, with a noticeable jump b/w the lower & higher values

Chart 2: Resting pulse n=40

Normal: Approximately yes

Shape: light-tailed

Evidence: points from 2 clusters with a noticeable jump b/w the lower & higher values.

Chart 3: plant height (cm) n=40

Normal: NO

Shape: left skewed

Evidence: They lower tail extends farther from the reference line, while upper values cluster near 44-45 cm

Chart 4: Resting Pulse

Normal: No

Shape: Normal with outliers

Evidence: Most points follow line, but a few extreme values lie far from it, especially the high value near 89 bpm.

Part B

B1: The distribution is left skewed. Apply a suitable power transformation, such as squaring the values, and then check Q-Q plot again before using mean and standard deviation. methods.

B2:
```
ggplot(df, aes(sample = value)) +
  stat_qq() +
  stat_qq_line()
```